Table 1.

Functional Suitability Test

Test on Module	Steps Undertaken
Login Module (Owner)	1. Opened the system login page. 2. Typed email: *owner@email.com*. 3. Typed password. 4. Clicked **Login** button. 5. Observed results.
Login Module (Manager)	1. Opened the system login page. 2. Typed email: *manager@email.com*. 3. Typed password. 4. Clicked **Login** button. 5. Observed results.
Login Module (Tenant)	1. Opened the system login page. 2. Typed email: *tenant@email.com*. 3. Typed password. 4. Clicked **Login** button. 5. Observed results.
Login Module (Invalid Credentials)	1. Opened the system login page. 2. Typed incorrect email and password. 3. Clicked **Login** button. 4. Observed results.
Password Reset Module	1. Clicked **Forgot Password** link on the login page. 2. Typed email: *owner@email.com*. 3. Clicked **Send Reset Link** button. 4. Observed results.
Logout Module	1. Logged in as any user role. 2. Clicked **Logout** button in the sidebar. 3. Observed results.
Owner Dashboard (Overview)	1. Logged in as Owner. 2. Navigated to **Overview** tab. 3. Selected the Month and Year filter. 4. Viewed statistics cards and recent activity history. 5. Observed results.

Property Management (Add Property)	1. Logged in as Owner. 2. Navigated to **Manage My Apartment** tab. 3. Clicked **Add Property** button. 4. Typed building name and filled in address fields. 5. Clicked **Save** button. 6. Observed results.
Unit Management (Add Unit)	1. Logged in as Owner. 2. Navigated to **Manage My Apartment** tab. 3. Selected a property: *Apartment 1*. 4. Clicked **Add Unit** button. 5. Typed unit name, monthly rent, and max occupancy. 6. Clicked **Save** button. 7. Observed results.
Unit Management (Edit Unit)	1. Logged in as Owner. 2. Clicked on a unit to open its detail modal. 3. Changed the monthly rent and unit status. 4. Clicked **Save** button. 5. Observed results.
User Management (Invite Manager)	1. Logged in as Owner. 2. Navigated to **User Management** tab then **Managers** sub-tab. 3. Clicked **Add Manager** button. 4. Filled in first name, last name, email, and phone number. 5. Clicked **Save** button. 6. Observed results.
User Management (Approve Tenant)	1. Logged in as Owner. 2. Navigated to **User Management** tab then **Tenants** sub-tab. 3. Filtered by *Pending Verification* status. 4. Clicked **Approve** on a pending tenant. 5. Observed results.
Maintenance Request (Owner View)	1. Logged in as Owner. 2. Navigated to **Maintenance Request** tab. 3. Applied status and priority filters. 4. Searched by keyword: *gripo*. 5. Observed results.
Payment Verification (Owner)	1. Logged in as Owner. 2. Navigated to **Payment History** tab. 3. Located a payment with pending verification status. 4. Reviewed the receipt image. 5. Clicked **Approve** button. 6. Observed results.

Analytics Module (Charts)	1. Logged in as Owner. 2. Navigated to **Analytics** tab. 3. Viewed the revenue line chart, property distribution pie chart, and maintenance trends bar chart. 4. Observed results.
Audit Reports Module (Quarterly Report)	1. Logged in as Owner. 2. Navigated to **Audit Reports** tab. 3. Selected Year: *2025* and Quarter: *Q1*. 4. Viewed summary cards and monthly breakdown. 5. Observed results.
Audit Reports Module (Add Expense)	1. Logged in as Owner. 2. Navigated to **Audit Reports** then Expenses section. 3. Clicked **Add Expense** button. 4. Filled in date, type, description, and amount. 5. Clicked **Save** button. 6. Observed results.
Activity Logs Module	1. Logged in as Owner. 2. Navigated to **Activity Logs** tab. 3. Applied role filter and date range: *06/01/2025 to 06/15/2025*. 4. Searched by keyword. 5. Observed results.
Announcements Module (Owner)	1. Logged in as Owner. 2. Clicked **New Announcement** button. 3. Typed title and message content. 4. Selected recipient tenants. 5. Clicked **Publish** button. 6. Observed results.
Account Settings (Owner)	1. Logged in as Owner. 2. Navigated to **Account Settings** tab. 3. Updated phone number and clicked **Save**. 4. Typed current password, new password, and confirm password. 5. Clicked **Submit** button. 6. Observed results.
Manager Dashboard (Overview)	1. Logged in as Manager. 2. Navigated to **Overview** tab. 3. Viewed statistics cards for active tenants, maintenance, and managed apartments. 4. Observed results.

Manage Apartment (Manager)	1. Logged in as Manager. 2. Navigated to **Manage Apartment** tab. 3. Viewed managed units with tenant assignments and status indicators. 4. Observed results.
Maintenance Request (Update Status)	1. Logged in as Manager. 2. Navigated to **Maintenance Request** tab. 3. Located a pending request. 4. Clicked **Mark as In Progress** button. 5. Observed results.
Payment Verification (Manager)	1. Logged in as Manager. 2. Navigated to **Payment History** tab. 3. Located a pending cash payment in the verification queue. 4. Reviewed receipt image and clicked **Approve**. 5. Observed results.
Record Cash Payment (Manager)	1. Logged in as Manager. 2. Clicked **Record Cash Payment** button. 3. Selected tenant, billing period, and payment mode. 4. Clicked **Submit** button. 5. Observed results.
Documents Module (Upload)	1. Logged in as Manager. 2. Navigated to **Documents** tab. 3. Selected a PDF file and assigned it to a tenant. 4. Clicked **Upload** button. 5. Observed results.
Notifications Module (Manager)	1. Logged in as Manager. 2. Navigated to **Notifications** tab. 3. Clicked on an unread notification to view details. 4. Clicked **Mark All as Read** button. 5. Observed results.
Announcements Module (Manager)	1. Logged in as Manager. 2. Clicked **New Announcement** button. 3. Typed title and message content. 4. Clicked **Publish** button. 5. Observed results.
Account Settings (Manager)	1. Logged in as Manager. 2. Navigated to **Account Settings** tab. 3. Updated phone number and birthdate. 4. Changed password and clicked **Submit**. 5. Observed results.

Tenant Dashboard (Overview)	1. Logged in as Tenant. 2. Navigated to **Overview** tab. 3. Viewed statistics, unit information, and lease details. 4. Scrolled through recent activity. 5. Observed results.
Maintenance Request (Submit Request)	1. Logged in as Tenant. 2. Navigated to **Request Maintenance** then **Request** tab. 3. Typed title and description of the issue. 4. Selected priority level: *High*. 5. Uploaded photos of the issue. 6. Clicked **Submit** button. 7. Observed results.
Maintenance Request (Track Request)	1. Logged in as Tenant. 2. Navigated to **Request Maintenance** then **Tracking** tab. 3. Viewed the list of submitted requests with status badges. 4. Clicked on a request to view full details. 5. Observed results.
Maintenance Request (Rate Service)	1. Logged in as Tenant. 2. Located a resolved maintenance request. 3. Selected a star rating and typed a review comment. 4. Clicked **Submit Review** button. 5. Observed results.
Payment Module (Submit Proof)	1. Logged in as Tenant. 2. Navigated to **Payments** tab. 3. Selected billing period and payment mode: *GCash*. 4. Uploaded receipt image. 5. Clicked **Submit** button. 6. Observed results.
Payment Module (Payment History)	1. Logged in as Tenant. 2. Navigated to **Payments** then **Payment History** section. 3. Viewed the list of all payments with status and mode. 4. Observed results.
Documents Module (Tenant View)	1. Logged in as Tenant. 2. Navigated to **Documents** tab. 3. Viewed the list of shared documents. 4. Clicked **Download** on a document. 5. Observed results.

Notifications Module (Tenant)	1. Logged in as Tenant. 2. Navigated to **Notifications** tab. 3. Checked the unread badge count. 4. Clicked on an unread notification to view full content. 5. Observed results.
Account Settings (Tenant)	1. Logged in as Tenant. 2. Navigated to **Account Settings** tab. 3. Updated phone number and birthdate. 4. Changed password and clicked **Submit**. 5. Observed results.
Occupant Management (Tenant)	1. Logged in as Tenant. 2. Navigated to **Account Settings** then **Occupants** section. 3. Clicked **Add Occupant** button. 4. Filled in name, sex, phone, and birthdate. 5. Clicked **Save** button. 6. Observed results.
Real-Time Updates (Maintenance)	1. Logged in as Tenant in Browser 1 and as Owner in Browser 2. 2. Tenant submitted a new maintenance request. 3. Switched to Owner's browser and checked the dashboard. 4. Observed results — request appeared without manual refresh.
Real-Time Updates (Payment)	1. Logged in as Tenant in Browser 1 and as Owner in Browser 2. 2. Tenant submitted a payment proof. 3. Switched to Owner's browser and checked the payment queue. 4. Observed results — payment appeared without manual refresh.
Real-Time Updates (Announcement)	1. Logged in as Owner in Browser 1 and as Tenant in Browser 2. 2. Owner published a new announcement. 3. Switched to Tenant's browser and checked the notifications tab. 4. Observed results — notification appeared without manual refresh.

Table 2.

Usability Test

Test on Module	Steps Undertaken
Login Module	1. Opened the system login page. 2. Typed username and password. 3. Clicked **Login** button. 4. Observed results.
Owner Dashboard (Overview Navigation)	1. Logged in as Owner. 2. Clicked **Overview** tab in the sidebar. 3. Scrolled through the dashboard statistics and recent activity. 4. Used the Month/Year filter dropdown. 5. Observed results.
Property Management (Add Property)	1. Logged in as Owner. 2. Clicked **Manage My Apartment** tab. 3. Clicked **Add Property** button. 4. Entered property name: *Sunrise Residences*. 5. Filled in the address fields. 6. Clicked **Save** button. 7. Observed results.
Unit Management (Add and Edit Unit)	1. Logged in as Owner. 2. Selected a property from the list. 3. Clicked **Add Unit** button. 4. Typed unit name: *Unit 5*, rent: *₱10,000*. 5. Clicked **Save** button. 6. Clicked on the newly added unit to edit its details. 7. Observed results.
User Management (Invite Manager)	1. Logged in as Owner. 2. Clicked **User Management** then **Managers** tab. 3. Clicked **+Add Manager** button. 4. Entered manager name: *Maria Santos*. 5. Typed email and phone number. 6. Clicked **Save** button. 7. Observed results.
Maintenance Request (Owner Viewing)	1. Logged in as Owner. 2. Clicked **Maintenance Request** tab. 3. Used the status filter dropdown to select *Pending*. 4. Typed in the search field to find a specific request. 5. Observed results.

Payment History (Verify Payment)	1. Logged in as Owner. 2. Clicked **Payment History** tab. 3. Located a pending payment in the verification queue. 4. Viewed the payment receipt image. 5. Clicked **Approve** button. 6. Observed results.
Audit Reports Module (Generate Report)	1. Logged in as Owner. 2. Clicked **Audit Reports** tab. 3. Selected Year: *2025*, Quarter: *Q2*. 4. Viewed summary cards and monthly revenue breakdown. 5. Observed results.
Activity Logs Module (Filter and Search)	1. Logged in as Owner. 2. Clicked **Activity Logs** tab. 3. Clicked the **Filter** button. 4. Selected role: *Tenant* and set date range. 5. Observed results.
Announcements Module (Publish Announcement)	1. Logged in as Owner. 2. Clicked **New Announcement** button. 3. Typed title: *Water Interruption Advisory*. 4. Typed the message body. 5. Selected recipients: *All Tenants*. 6. Clicked **Publish** button. 7. Observed results.
Manager Dashboard (Overview Navigation)	1. Logged in as Manager. 2. Clicked **Overview** tab in the sidebar. 3. Viewed statistics cards and recent maintenance/payment activity. 4. Observed results.
Maintenance Request (Manager Status Update)	1. Logged in as Manager. 2. Clicked **Maintenance Request** tab. 3. Located a pending request from a tenant. 4. Clicked **Mark as In Progress** button. 5. Observed results.
Record Cash Payment (Manager)	1. Logged in as Manager. 2. Clicked **Payment History** tab. 3. Clicked **Record Cash Payment** button. 4. Selected tenant: *Leo Borromeo* and billing period. 5. Clicked **Submit** button. 6. Observed results.

Documents Module (Upload Document)	1. Logged in as Manager. 2. Clicked **Documents** tab. 3. Selected a PDF file to upload. 4. Assigned it to tenant: *Leo Borromeo*. 5. Clicked **Upload** button. 6. Observed results.
Tenant Dashboard (Overview Navigation)	1. Logged in as Tenant. 2. Clicked **Overview** tab in the sidebar. 3. Viewed statistics cards, unit information, and lease dates. 4. Scrolled through recent activity and FAQ section. 5. Observed results.
Submit Maintenance Request (Tenant)	1. Logged in as Tenant. 2. Clicked **Request Maintenance** tab. 3. Typed title: *Sira ang gripo sa kusina*. 4. Typed description of the issue. 5. Selected priority: *High*. 6. Uploaded photos using the photo slots. 7. Clicked **Submit** button. 8. Observed results.
Submit Payment Proof (Tenant)	1. Logged in as Tenant. 2. Clicked **Payments** tab. 3. Selected billing period: *June 2025*. 4. Selected payment mode: *GCash*. 5. Uploaded receipt image. 6. Clicked **Submit** button. 7. Observed results.
View Documents (Tenant)	1. Logged in as Tenant. 2. Clicked **Documents** tab. 3. Viewed the list of shared documents. 4. Clicked **Download** button on a document. 5. Observed results.
Account Settings (Update Profile)	1. Logged in as Tenant. 2. Clicked **Account Settings** tab. 3. Updated phone number: *+639201234567*. 4. Clicked **Save** button. 5. Changed password using the password form. 6. Observed results.

Manage Occupants (Add Occupant)	1. Logged in as Tenant. 2. Clicked **Account Settings** then **Occupants** section. 3. Clicked **Add Occupant** button. 4. Entered occupant name: *Maria Borromeo*. 5. Filled in sex, phone, and birthdate fields. 6. Clicked **Save** button. 7. Observed results.

Table 3.

Performance Efficiency Test

Test on Module	Steps Undertaken
Login Module (Response Time)	1. Opened the system login page. 2. Typed valid email and password. 3. Clicked **Login** button. 4. Measured the time from button click to dashboard load. 5. Observed results.
Dashboard Loading (Owner)	1. Logged in as Owner. 2. Navigated to the **Overview** tab. 3. Measured the time for statistics cards and recent activity to fully load. 4. Observed results.
Dashboard Loading (Manager)	1. Logged in as Manager. 2. Navigated to the **Overview** tab. 3. Measured the time for statistics cards and recent activity to fully load. 4. Observed results.
Dashboard Loading (Tenant)	1. Logged in as Tenant. 2. Navigated to the **Overview** tab. 3. Measured the time for statistics cards, unit info, and lease details to fully load. 4. Observed results.
Property Management (Load Units List)	1. Logged in as Owner. 2. Navigated to **Manage My Apartment** tab. 3. Selected a property with multiple units. 4. Measured the time for all units to render in the list. 5. Observed results.
Maintenance Request (Load Request List)	1. Logged in as Owner. 2. Navigated to **Maintenance Request** tab. 3. Measured the time for the request list with status badges to fully load. 4. Applied a filter and measured the re-render time. 5. Observed results.

Payment History (Load Records)	1. Logged in as Owner. 2. Navigated to **Payment History** tab. 3. Measured the time for payment records and statistics to load. 4. Applied status filter and measured the re-render time. 5. Observed results.
Analytics Module (Chart Rendering)	1. Logged in as Owner. 2. Navigated to **Analytics** tab. 3. Measured the time for all charts (revenue, property, maintenance) to fully render. 4. Observed results.
Audit Reports (Generate Report)	1. Logged in as Owner. 2. Navigated to **Audit Reports** tab. 3. Selected Year: *2025* and Quarter: *Q1*. 4. Measured the time for the report data and summary cards to load. 5. Observed results.
Activity Logs (Load and Filter)	1. Logged in as Owner. 2. Navigated to **Activity Logs** tab. 3. Measured the time for the log entries to load. 4. Applied a date range filter and measured the re-render time. 5. Observed results.
Real-Time Update (Propagation Speed)	1. Logged in as Tenant in Browser 1 and as Owner in Browser 2. 2. Tenant submitted a new maintenance request. 3. Measured the time from submission to the request appearing on Owner's dashboard. 4. Observed results.
Payment Submission (Upload Speed)	1. Logged in as Tenant. 2. Navigated to **Payments** tab. 3. Selected billing period and payment mode. 4. Uploaded a receipt image and clicked **Submit**. 5. Measured the time from submit to confirmation display. 6. Observed results.
Maintenance Photo Upload (Upload Speed)	1. Logged in as Tenant. 2. Navigated to **Request Maintenance** tab. 3. Uploaded 4 photos using the photo slots. 4. Measured the time for each photo to upload and display the preview. 5. Observed results.

Search and Filter (Response Time)	1. Logged in as Owner. 2. Navigated to **Maintenance Request** tab. 3. Typed a keyword in the search field. 4. Measured the time for filtered results to appear. 5. Observed results.
Page Navigation (Tab Switching Speed)	1. Logged in as Owner. 2. Clicked through each sidebar tab: Overview, Manage Apartment, User Management, Maintenance, Payments, Analytics, Audit Reports, Activity Logs, Account Settings. 3. Measured the transition time between each tab. 4. Observed results.